Understanding Sensitivity Analyses

What is a Sensitivity Analysis?

A sensitivity analysis is a method used to determine how changes in variables impact an outcome. In the context of Operations Research, it is a method that is used to determine how changes in parameters (like cost, demand, available resources) impact the optimal solution of an optimization model.

To perform a sensitivity analysis, a baseline model is defined and solved. Key parameters are then identified and tested one at a time, such as increasing or decreasing demand by 10%. After each change, the model is resolved and the results are compared to the baseline solution. This process helps identify which parameters have the greatest influence on the model's performance, and helps guide future decisions.

How Do Changes in Parameters Affect an Optimization Model?

Optimization models rely on parameter data to determine the best solution. Changes in parameters can alter the objective value, decision variables, and structure of the optimal solution.

Consider a furniture manufacturing company that uses an optimization model to determine how many chairs and tables to produce each week to maximize profit. The model uses parameters such as customer demand, material costs, labor availability, and selling prices.

If customer demand for tables increases, the model may recommend the company to produce more tables and fewer chairs to maximize profit. If the cost of lumber increases significantly, the model may shift production towards products that use less wood. If labor hours become limited because of worker shortages, the model may prioritize products with the highest profit per labor hour.

In this example, changes in parameters affect the optimal production plan and the profit earned by the company. By analyzing how the solution changes when parameters are modified, the company can identify which factors have the greatest impact on profit and make the right decisions when conditions change.

Sensitivity Analysis Using AMPL

AMPL is a programming language used to model and solve optimization problems. When AMPL is used with a solver such as Gurobi, it can provide sensitivity analysis information that helps users understand how the optimal solution responds to changes in parameters.

The following are Gurobi suffixes that can be used in AMPL to extract sensitivity analysis results.

.slack:
This suffix shows the amount of unused capacity in a constraint at the optimal solution. In the furniture manufacturing example, .slack could indicate how many labor hours or units of raw material are unused in the optimal solution.

.dual:
This suffix shows how much the objective value would change if the available amount of a resource increased by one unit. In the furniture manufacturing problem, the dual value would indicate how much additional profit the company would gain from one additional hour of labor.

.rc (reduced cost):
This suffix applies to variables that are not in the optimal solution. It shows how much the profit/cost coefficient must improve before the variable becomes profitable enough to enter the optimal solution. In the furniture manufacturing example, if a product is not being produced, .rc shows how much its profit per unit would need to increase before the model would recommend producing it.

What are the Real-World Applications of Sensitivity Analysis?

Sensitivity analysis is widely used in real-world decision-making where conditions are always changing. It helps organizations understand risk and prepare for different scenarios. It can be applied across many fields including: supply chain management, healthcare resource allocation, airline scheduling, financial planning, energy pricing, and more. Here's a closer look.

Supply Chain Management:
Suppose a resale company relies on overseas suppliers for raw materials, but shipping delays or geopolitical issues disrupt supply. A sensitivity analysis can show how reduced material availability affects production levels and profit, helping the company decide how much backup inventory to hold or whether to look for new suppliers.

Airline Scheduling:
Airlines optimize flight schedules based on demand, aircraft availability, and fuel costs. Since fuel costs fluctuate, a sensitivity analysis can help determine how much a rise in fuel cost will affect profit and whether certain routes should be reduced or canceled.

Energy Pricing
Energy companies generate energy based on demand forecasts, fuel costs, and regulatory constraints. A sensitivity analysis helps evaluate how changes in demand or fuel costs affect production plans and costs, supporting more stable pricing.